

MELSOM, NorwayWMO: **014810**Lat: **59.230N**Lon: **10.348E**Elev: **85**StdP: **14.65**Time Zone: **1.00 (EUC)**Period: **04-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.6	9.6	0.5	5.6	6.6	5.0	7.1	11.5	16.7	42.0	15.0	41.5	2.5	0	0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.3	79.2	63.3	75.6	62.0	72.2	60.6	65.7	75.0	64.0	72.1	62.4	69.3	4.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.5	84.7	68.4	60.9	80.0	66.7	59.4	75.9	65.3	30.5	75.5	29.3	72.6	28.1	69.3	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.5	11.7	10.1	0.7	83.1	5.8	4.0	-3.4	85.9	-6.8	88.2	-10.1	90.5	-14.3	93.3		
(5)			0.4	68.0	5.7	2.1	-3.7	69.5	-7.0	70.8	-10.2	71.9	-14.3	73.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.1	28.5	29.1	34.6	43.2	52.2	59.1	63.2	61.2	54.8	45.7	37.3	31.0
	DBStd	14.03	9.01	8.45	7.69	5.53	6.40	4.53	4.55	3.75	4.62	5.74	6.88	9.72
	HDD50	3130	665	584	478	213	46	0	0	0	14	157	383	590
	HDD65	7325	1130	1004	941	653	400	187	91	128	306	598	832	1055
	CDD50	1339	0	0	2	11	114	272	410	348	157	24	1	0
	CDD65	58	0	0	0	0	3	9	36	10	0	0	0	0
	CDH74	479	0	0	0	0	37	116	268	58	0	0	0	0
	CDH80	77	0	0	0	0	2	15	57	3	0	0	0	0
Wind	WSAvg	4.6	5.1	5.0	4.9	4.8	4.7	4.3	3.9	3.9	4.2	4.6	5.3	5.1
Precipitation	PrecAvg	38.3	3.3	2.4	1.9	2.1	2.5	2.7	3.1	4.6	3.6	4.4	4.4	3.3
	PrecMax	53.7	7.4	7.3	3.9	4.5	6.0	5.2	6.9	8.9	11.1	12.3	8.3	7.2
	PrecMin	28.6	0.6	0.3	0.0	0.6	0.1	0.2	0.5	1.3	1.5	0.3	0.7	1.1
	PrecStd	7.0	1.8	1.7	1.2	1.1	1.5	1.2	1.4	1.9	2.4	2.8	1.9	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.4	50.1	60.1	65.6	78.7	81.9	85.8	79.3	70.1	61.0	54.4	50.3
		MCWB	43.3	42.0	48.0	51.0	60.4	62.0	66.0	65.5	59.4	53.8	50.7	47.1
	2%	DB	45.0	45.2	53.4	60.5	72.9	77.6	80.9	74.8	67.2	58.3	52.3	48.1
		MCWB	42.1	39.3	44.1	48.5	58.1	61.1	64.9	62.7	57.9	53.4	49.1	45.4
	5%	DB	42.9	42.1	48.0	56.8	68.2	72.6	76.6	71.6	65.0	56.5	50.2	46.2
		MCWB	40.2	38.5	40.4	46.9	56.0	58.9	63.1	61.3	56.8	52.7	47.7	43.6
	10%	DB	40.7	40.1	44.7	53.7	63.9	69.0	73.0	69.2	62.7	54.8	47.8	43.9
		MCWB	37.9	37.6	38.8	45.3	53.7	57.0	62.1	60.5	56.4	51.5	45.3	41.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.8	43.7	48.8	53.4	63.1	65.4	68.1	67.0	62.6	57.0	52.9	48.3
		MCDB	46.4	47.6	59.1	60.5	73.7	75.5	79.2	74.1	65.8	58.9	53.7	49.8
	2%	WB	42.8	40.8	44.5	50.3	60.2	62.9	66.4	64.8	60.6	55.2	49.5	46.0
		MCDB	44.5	43.3	51.9	57.9	69.2	74.0	77.2	71.5	64.2	57.1	50.9	47.7
	5%	WB	40.7	39.3	42.0	48.4	57.7	60.8	64.9	63.0	59.2	53.8	47.5	44.0
		MCDB	42.5	41.4	46.2	55.5	66.1	69.6	74.2	69.0	62.7	56.0	49.2	45.8
	10%	WB	38.3	37.9	40.1	46.3	55.2	59.0	63.2	61.6	57.7	52.0	45.4	41.6
		MCDB	40.4	39.9	43.7	52.3	61.9	66.2	70.7	66.9	61.4	54.4	47.4	43.5
Mean Daily Temperature Range	5% DB	MDBR	9.2	10.3	13.9	15.0	15.6	16.0	15.3	14.8	13.8	11.1	9.2	9.4
		MCDBR	10.3	12.9	19.1	20.4	22.2	22.5	21.4	18.4	16.8	12.8	11.5	10.0
		MCWBR	9.6	9.3	11.7	11.2	10.5	10.5	9.0	9.0	10.0	9.3	10.4	9.4
	5% WB	MCDBR	9.8	10.3	16.2	18.3	19.1	18.5	18.6	15.1	13.4	10.4	10.5	9.8
		MCWBR	9.6	8.1	10.7	10.7	9.7	10.0	8.9	8.5	9.4	8.9	10.3	9.7
Clear-Sky Solar Irradiance	taub	0.280	0.317	0.337	0.377	0.371	0.358	0.380	0.376	0.346	0.333	0.298	0.256	
	taud	2.233	2.309	2.355	2.312	2.410	2.468	2.427	2.434	2.505	2.437	2.293	2.158	
	Ebn at Noon	167	221	254	261	272	278	268	260	251	215	159	131	
	Edh at Noon	14	22	28	34	33	32	32	30	24	19	13	10	
All-Sky Solar Radiation	RadAvg	99	287	764	1223	1614	1842	1650	1297	860	384	133	68	
	RadStd	19	54	78	169	163	123	195	90	96	68	33	7	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (54 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A